



Bahir Dar University
Faculty of computing department of software
engineering
Operating system and system programming
Group Assignment(Group 2)

Member's Name	Student ID
1 Adane Birhanu	BDU1701496
2 Abel Eyayu.....	BDU1701445
3 Eyob Abay	BDU1701937
4 Endawek Basaznew	BDU1701024
5 Rahel Demissie	BDU1702504

SUBMITTED TO: : Lec. Wondmu
SUBMISSION DATE: 28/04/2026 G.C

Introduction	3
1. AMD – Processor Manufacturer	4
1.1 Introduction	4
1.2 History	5
1.3 Objectives / Core Focus	6
1.4 Key Characteristics	7
1.5 AMD Products & Technologies	8
2. HP Inc(CSM / OEM)	9
2.1 Introduction	9
2.2 History	10
2.3 Objectives	10
2.4 Key Characteristics	11
2.5 Products & Technologies	13
3.Nanya Technology	15
3.1 Introduction	15
3.2 History	15
3.3 Objectives	16
3.4 Key Characteristics	17
3.4 Products & Technologies	18
4.Kioxia(SSD / NAND Flash)	20
4.1 Introduction	20
4.2 History	20
4.3 Objectives / Core Focus	22
4.4 Key Characteristics	23
4.5 Products & Technologies	24
5.Market Context & Industry Trends	26
6.Risks & Challenges	29
7.Sustainability & Future Outlook	31
Conclusion	34
REFERENCE	35

Introduction

Building a modern laptop or desktop PC is never the work of a single company. It's the result of a sophisticated, tightly coordinated supply chain where each specialist brings their unique expertise to create something greater than the sum of its parts. In 2026, this collaboration has become even more vital as the industry shifts rapidly toward intelligent, AI-powered computing.

At the center of one such powerful ecosystem are four key players:

- **AMD** creates the high-performance processors that act as the brain of the system — handling complex calculations, stunning graphics, multitasking, and the growing demands of on-device artificial intelligence.
- **HP Inc.** serves as the system maker and integrator. It carefully selects components, designs the motherboard, optimizes thermal and power solutions, develops firmware, runs exhaustive testing, and delivers reliable, beautifully engineered laptops and desktops that millions of people trust every day.
- **Nanya Technology** supplies the fast DRAM (RAM) — the high-speed working memory that allows the processor to juggle multiple applications, load data quickly, and run memory-intensive AI features without hesitation.
- **Kioxia** provides the advanced NAND flash memory chips that form the foundation of fast, durable solid-state drives (SSDs), delivering quick boot times, responsive file access, and dependable long-term storage.

Imagine opening a sleek new HP laptop on a typical workday. It wakes instantly, smoothly handles video calls with AI noise cancellation and background enhancement, edits large documents or photos without lag, runs local language models for instant summarization, and still delivers excellent battery life. That seamless, responsive experience doesn't happen by accident. It emerges from the close partnership between AMD's intelligent processors, Nanya's responsive memory, Kioxia's high-performance storage, and HP's expert system integration.

In today's dynamic market, this collaboration is more important than ever. The rapid adoption of **AI PCs** is raising expectations for performance, efficiency, and intelligence while putting pressure on memory and storage supply. Every link in the chain must deliver not just higher specifications, but also better power efficiency and long-term reliability.

By understanding these four companies and how they interconnect, engineers gain insight into optimization opportunities, procurement teams make smarter sourcing decisions, product managers appreciate the challenges of qualification and integration, and technology enthusiasts develop a deeper respect for the engineering behind the devices they use daily.

Behind every reliable HP system lies months of careful planning, rigorous testing, and genuine technological synergy—a quiet but impressive example of what focused collaboration can achieve in the fast-moving world of personal computing.

1. AMD – Processor Manufacturer

1.1 Introduction

At the heart of every high-performance laptop and desktop lies its processor — the component that acts as the brain, the decision-maker, and increasingly, the intelligent core of the entire system. This is where AMD shines.

AMD (Advanced Micro Devices, Inc.) is a fabless semiconductor company that designs powerful central processing units (CPUs), graphics processing units (GPUs), and specialized AI accelerators. In the PC supply chain, AMD serves as the primary processor manufacturer, creating the main compute engine that powers everything from everyday productivity tasks to demanding creative workloads and advanced on-device artificial intelligence.

When you open a new HP laptop and experience smooth multitasking, crisp graphics, fast application launches, or smart AI features like real-time noise cancellation during video calls and instant document summarization, much of that responsive feel comes directly from AMD's Ryzen processors working efficiently beneath the surface.

In 2026, AMD has become one of the driving forces behind the AI PC revolution. Its latest Ryzen AI processors are not just faster — they are fundamentally smarter, equipped with powerful Neural Processing Units (NPUs) that allow computers to handle complex AI tasks locally, right on the device, delivering better privacy, lower latency, and improved battery efficiency.

What makes AMD particularly valuable in the partnership with HP is its ability to deliver a compelling balance of high performance, strong multi-core capabilities, excellent power efficiency, and outstanding value. This combination enables HP to offer laptops and desktops that feel premium without becoming prohibitively expensive, giving users noticeably better everyday experiences.

This version is warmer, more engaging, and flows naturally while staying professional. It connects AMD's technology directly to the user's real-world experience with HP devices and highlights its growing importance in the AI era.

1.2 History

AMD's story is one of resilience, reinvention, and remarkable comeback — a true Silicon Valley underdog tale that continues to inspire the tech industry.

Founded in 1969 in Sunnyvale, California, by a group of engineers who had previously worked at Fairchild Semiconductor, Advanced Micro Devices began as a scrappy startup aiming to provide alternative sources for logic and memory chips. For its first several decades, AMD operated primarily as a second-source supplier in an industry dominated by much larger and more powerful competitors. While it earned a reputation for solid engineering, it remained largely in the shadow of its bigger rival for many years.

The turning point came in the late 2010s. After years of heavy investment in research and development, AMD launched the first-generation Ryzen processor family in 2017. These chips marked a dramatic breakthrough, delivering superior multi-core performance and far better value than anything previously available. For the first time in decades, AMD wasn't just competing — it was winning hearts and minds among

gamers, content creators, and everyday users who finally felt they were getting more performance for their money.

Under the calm, visionary leadership of CEO Dr. Lisa Su, who took the helm in 2014, AMD executed one of the most impressive corporate turnarounds in technology history. The company shifted from a “me-too” supplier to a genuine innovator by embracing a bold chiplet-based design philosophy, investing heavily in advanced manufacturing partnerships, and fostering a culture of speed and engineering excellence.

From near-bankruptcy fears in the early 2010s to becoming a major force in both consumer and enterprise computing, AMD’s journey reflects the power of persistent innovation and strategic focus. By 2026, the company has firmly established itself as one of the most respected names in high-performance computing, particularly in the rapidly growing field of AI-accelerated processors.

1.3 Objectives / Core Focus

AMD’s core mission is clear and ambitious: to deliver high-performance computing experiences that push the boundaries of what’s possible, while never compromising on power efficiency and real-world usability.

In today’s fast-evolving landscape, AMD is deeply committed to shaping the future of personal computing through the AI PC era. The company is not just focused on making processors faster it is fundamentally redefining how intelligent a personal computer can be. By integrating powerful Neural Processing Units (NPUs) into its Ryzen processors, AMD aims to enable rich, on-device artificial intelligence capabilities that run locally with high speed, strong privacy, and excellent energy efficiency.

AMD strives to create processors that excel across diverse user needs — whether it’s a student juggling multiple applications and browser tabs, a content creator editing high-resolution videos, a gamer seeking smooth frame rates, or a professional relying on secure and responsive AI tools for daily work. The goal is to make advanced computing feel effortless and accessible, not just for power users, but for everyone.

A key part of AMD's philosophy is close collaboration with system makers like HP. Rather than simply selling chips, AMD works hand-in-hand with HP's engineering teams to co-optimize platforms — fine-tuning everything from power delivery and thermal management to firmware and AI software integration. This partnership-driven approach ensures that the final HP laptops and desktops deliver balanced, reliable, and truly impressive user experiences.

1.4 Key Characteristics

Advanced Micro Devices isn't just a chip company—it's an engineering-driven organization that focuses on making powerful computing more accessible and efficient.

What Defines AMD

1. Performance that scales with your workload

AMD chips are known for handling multiple tasks extremely well. Whether it's gaming, video editing, or running AI models, their processors are built to keep things smooth even under heavy load.

2. Smart design philosophy (chiplets)

Instead of building one large chip, AMD often uses smaller pieces ("chiplets") combined together. This approach improves efficiency, reduces cost, and allows faster innovation compared to traditional designs.

3. Strong value for money

AMD has built a reputation for offering high performance at competitive prices. This makes their processors attractive not only for enthusiasts but also for large OEMs like HP Inc..

4. AI is becoming a core focus

Modern AMD processors now include dedicated AI engines (NPUs). This means tasks like voice recognition, real-time translation, and AI assistants can run directly on your device without relying heavily on the cloud.

5. Balanced power efficiency

AMD aims to deliver strong performance without draining too much power—especially important for laptops where battery life matters.

6. Engineering-led culture

Under CEO Lisa Su, AMD has become highly execution-focused. The company is known for disciplined roadmaps and delivering on long-term technical goals.

7. Competitive and agile mindset

AMD thrives as a challenger. Its culture encourages fast innovation and bold moves, which helped it regain market share and push the industry forward.

Ultimately, AMD wants to be the engine that powers smarter, more efficient, and more capable computers — helping users accomplish more while using less energy and enjoying a smoother daily interaction with technology.

1.5 AMD Products & Technologies

At Advanced Micro Devices, products are designed as a single integrated computing platform, where CPU, GPU, and AI engines work together rather than as separate parts.

The Ryzen series forms the core of client computing, powering everyday laptops and PCs with strong multitasking performance. It also includes Ryzen PRO for enterprise use with added security and stability, and Ryzen AI processors, which introduce dedicated NPUs for on-device AI tasks such as transcription, assistance, and productivity features reducing dependence on the cloud.

Alongside this, Radeon graphics handle visual and parallel workloads like gaming, content creation, and rendering. In many systems, AMD combines CPU and GPU into a single APU design, improving efficiency and power usage.

For large-scale computing, EPYC processors extend AMD's architecture into servers and data centers, supporting AI, cloud services, and enterprise workloads, while sharing design principles with Ryzen to maintain consistency across platforms.

These products are unified through key technologies such as chiplet architecture, which improves scalability and manufacturing efficiency, and 3D V-Cache, which enhances performance in data-heavy tasks like gaming and simulation. Increasingly, AI acceleration through NPUs is becoming central, enabling faster and more efficient local AI processing.

In real systems built by partners like HP Inc., AMD's CPU, GPU, and AI engine work together with memory and storage to deliver a balanced, high-performance computing experience.

2. HP Inc(CSM / OEM)

2.1 Introduction

HP Inc. operates as a **Computer System Manufacturer (CSM)**, also commonly referred to as an **Original Equipment Manufacturer (OEM)**. In this role, HP is responsible for transforming individual electronic components into complete, fully functional computing systems such as laptops, desktops, and workstations.

Unlike semiconductor companies that design chips, HP's core strength lies in **system-level engineering and integration**. It carefully selects components—such as processors from Advanced Micro Devices, memory from Nanya Technology, and storage technologies enabled by Kioxia—and brings them together into a unified platform.

As a CSM/OEM, HP's responsibilities go far beyond assembly. The company designs system architecture, develops custom motherboards, optimizes thermal and power performance, builds firmware and BIOS, and conducts extensive validation testing to ensure reliability, compatibility, and long-term stability. Every component must meet strict qualification standards before it becomes part of a finished HP product.

This role positions HP at the **center of the PC supply chain**, acting as the bridge between component manufacturers and end users. By integrating hardware and software into a seamless experience, HP ensures that advanced technologies are delivered in a form that is practical, secure, and ready for real-world use.

In today's AI-driven computing era, HP's role as a system integrator is becoming even more critical. The company not only assembles devices but also optimizes entire platforms to support emerging workloads such as on-device artificial intelligence, hybrid work environments, and high-performance computing—making it a key enabler of modern PC innovation.

2.2 History

HP Inc. traces its origins back to 1939, when Bill Hewlett and Dave Packard founded the company in a small garage in Palo Alto. That garage later became widely recognized as the birthplace of Silicon Valley. In its early years, the company focused on electronic test and measurement instruments, building a reputation for engineering precision and innovation.

Over the decades, HP expanded into computing, becoming one of the pioneers of personal computers, printers, and enterprise technology solutions. Through the 1980s and 1990s, it established itself as a global leader in both hardware and IT infrastructure, known for reliability and strong engineering culture.

A major turning point came in 2015, when the original Hewlett-Packard Company split into two separate entities:

- HP Inc. – focused on personal systems (PCs) and printing
- Hewlett Packard Enterprise (**HPE**) – focused on enterprise servers, cloud, and services

Following the split, HP Inc. sharpened its focus on consumer and commercial computing devices, strengthening its position as one of the world's leading PC manufacturers. It continued to evolve its product portfolio across laptops, desktops, and workstations, while also building one of the most efficient global supply chains in the industry.

Today, HP's long history of engineering excellence and operational discipline supports its role as a **modern CSM/OEM**, enabling it to integrate advanced technologies—such as processors from Advanced Micro Devices—into reliable, high-performance systems used worldwide.

2.3 Objectives

At its core, HP Inc. aims to **translate advanced technology into practical, reliable, and user-friendly computing experiences**. As a Computer System Manufacturer (CSM/OEM), its objectives are centered not on creating individual components, but on delivering complete systems that perform consistently in real-world environments.

A primary focus for HP is **system reliability and quality**. Every device is designed to operate smoothly over long periods, whether in consumer use or demanding enterprise environments. This requires strict component selection, deep validation processes, and continuous optimization of hardware and software.

Another key objective is **seamless integration**. HP works to ensure that processors from Advanced Micro Devices, memory from Nanya Technology, and storage technologies enabled by Kioxia function together efficiently as a unified system. The goal is to deliver performance that feels smooth and balanced, rather than just powerful on paper.

HP also places strong emphasis on **security and manageability**, particularly for business customers. Its systems are designed with built-in protections at the hardware and firmware level, along with tools that allow organizations to manage devices at scale.

In 2026, a growing priority is **AI-driven computing**. HP is focused on building systems that can support on-device artificial intelligence, enabling features such as intelligent collaboration, automation, and enhanced productivity without relying heavily on cloud resources.

Finally, HP is committed to **sustainability and responsible innovation**. This includes reducing environmental impact through energy-efficient designs, sustainable materials, and more responsible manufacturing and supply chain practices.

Overall, HP's objective is to act as a **bridge between advanced semiconductor innovation and real-world usability**, ensuring that powerful technologies are delivered in a form that is dependable, secure, and accessible to users worldwide.

2.4 Key Characteristics

HP Inc. is fundamentally a **system integration company**, meaning its strength lies in how it brings together different technologies into a complete and reliable computer system. Unlike semiconductor firms, HP does not design core chips, but instead focuses on building fully functional devices from those components.

1. System Integration Capability

HP's most important characteristic is its ability to integrate components into a unified system. It combines processors from Advanced Micro Devices, memory from Nanya Technology, and storage from Kioxia, and ensures they all work together smoothly. This includes designing the motherboard, tuning firmware, and optimizing system performance so that the final product feels balanced and efficient rather than fragmented.

2. Focus on Reliability and Quality

Another key characteristic of HP is its strong emphasis on reliability. Every system goes through strict testing and validation before reaching the market. HP prioritizes long-term stability, meaning its devices are designed not just for peak performance, but for consistent operation over time. This is especially important for business users who depend on systems for critical workloads.

3. Strong Supply Chain and Execution Power

HP operates a highly efficient global supply chain that allows it to produce and distribute computers at massive scale. It manages sourcing, manufacturing coordination, logistics, and product delivery across different regions. This execution strength ensures that HP can meet global demand while maintaining cost efficiency and availability.

4. Security and Enterprise Readiness

A major part of HP's identity is its focus on security, especially for business environments. Its systems include hardware-level protections, secure firmware features, and management tools that allow IT teams to control and protect devices across large organizations. This makes HP a trusted choice in enterprise, education, and government sectors.

5. Balanced Engineering Approach

HP follows a design philosophy centered on balance rather than extreme performance alone. Each product is engineered to maintain equilibrium between performance, power efficiency, thermal control, and physical design. This ensures that users get a smooth experience without issues like overheating or excessive battery drain.

6. Transition Toward AI-Ready Computing

In recent years, HP has been shifting toward AI-ready systems. With the rise of processors from Advanced Micro Devices that include AI capabilities, HP is optimizing its systems to support features like on-device AI assistants, intelligent collaboration tools, and productivity enhancements that run locally instead of relying fully on the cloud.

2.5 Products & Technologies

HP Inc. builds its product and technology ecosystem around one core idea: **turning advanced components into complete, reliable computing systems for different types of users**. Instead of focusing on a single category, HP operates across consumer, business, and high-performance computing segments.

1. Consumer Computing Systems

HP's consumer products are designed for everyday users who need a balance of performance, design, and affordability. These include lines like Pavilion and Envy, which are built for tasks such as browsing, studying, entertainment, and light content creation. In these systems, HP integrates processors from Advanced Micro Devices along with memory from Nanya Technology and storage solutions powered by Kioxia, ensuring smooth and responsive daily performance.

2. Business and Enterprise Systems

For corporate environments, HP offers EliteBook and ProBook series, which are designed with a strong focus on security, stability, and manageability. These systems are built for long-term use in organizations where reliability is critical. HP adds enterprise-grade firmware, security layers, and IT management tools on top of

standard hardware, making these devices suitable for large-scale deployment across companies and institutions.

3. Gaming and High-Performance Systems

HP's OMEN series targets gamers and performance users who require higher graphics capability and thermal efficiency. These systems are designed to handle demanding workloads such as gaming, streaming, and creative production. HP engineers optimize cooling systems, power delivery, and display performance to maintain stable high performance over long sessions.

4. Core Hardware Integration Technologies

A key part of HP's technology strength is its system-level engineering. It does not manufacture chips, but it designs and integrates all components into a unified platform. This includes motherboard design, BIOS/firmware development, thermal engineering, and system optimization. The goal is to ensure that CPU, GPU, memory, and storage all function together efficiently as one system.

5. Storage, Memory, and Platform Coordination

HP systems rely on tightly coordinated hardware ecosystems. Memory modules from Nanya Technology provide fast multitasking capability, while NAND-based SSDs from Kioxia ensure fast boot times and data access. HP integrates these components with processors from Advanced Micro Devices to create balanced performance across workloads.

6. AI-Ready System Design (2026 Direction)

In the modern computing era, HP is increasingly focused on AI-enabled PCs. These systems are optimized for processors with built-in NPUs, allowing features such as real-time transcription, intelligent collaboration tools, and local AI processing. This reduces reliance on cloud computing and improves speed, privacy, and efficiency.

3.Nanya Technology

3.1 Introduction

Nanya Technology is a **semiconductor company that specializes in producing DRAM (Dynamic Random Access Memory)**, which is the short-term working memory used inside computers. In simple terms, it is one of the key companies that supplies the “active workspace” that allows a computer to run multiple programs smoothly at the same time.

Unlike companies that build processors or finished computers, Nanya focuses specifically on memory chips. These chips do not store data permanently, but instead temporarily hold information that the processor needs instantly while tasks are running. This makes DRAM one of the most important components for system speed and responsiveness.

In the PC ecosystem, Nanya plays the role of a **core memory supplier**, providing DRAM modules that are integrated into laptops and desktops by system manufacturers like HP Inc.. When combined with processors from Advanced Micro Devices and storage solutions from Kioxia, Nanya’s memory helps ensure that systems can multitask efficiently, handle modern applications, and support AI-driven workloads.

In a more human way, Nanya can be thought of as the company that provides a computer with its **short-term thinking space**—the faster and more reliable this memory is, the smoother the entire system feels when you open apps, switch tasks, or run demanding software.

3.2 History

Nanya Technology was established in **1995 in Taiwan**, as part of the larger Formosa Plastics Group. From the beginning, the company was created with a clear purpose: to become a **specialized manufacturer of DRAM memory chips**, focusing only on dynamic random access memory rather than spreading into multiple semiconductor fields.

In its early development phase, Nanya relied heavily on technology licensing and partnerships with established global memory leaders. This helped the company quickly enter the DRAM industry and build manufacturing capability at scale. Over time, it invested significantly in its own fabrication facilities (fabs) in Taiwan, gradually moving from a technology follower to an independent memory producer.

During the 2000s, Nanya experienced the typical cycles of the global DRAM industry—periods of strong growth followed by sharp price declines. These cycles shaped the company’s strategy, pushing it to focus more on **process efficiency, cost control, and stable production quality** rather than aggressive expansion.

In the 2010s, Nanya began transitioning toward more advanced DRAM generations, including DDR3 and later DDR4, strengthening its position as a reliable mid-tier supplier in the global memory market. While it did not compete directly with the largest DRAM giants in scale, it built a strong reputation for **consistent supply and engineering stability**.

By the early 2020s and into 2026, Nanya had firmly positioned itself as a **Tier-2 global DRAM supplier**, serving PC manufacturers such as HP Inc. and other electronics companies. The company benefited from rising demand for memory in AI PCs and data-intensive computing systems, along with tighter global supply conditions.

A major recent milestone in its history came in 2026, when strategic interest in DRAM supply security increased globally, leading to stronger industry partnerships and investments involving memory manufacturers, including collaboration trends with companies like Kioxia in the broader memory ecosystem.

Today, Nanya’s history reflects a steady evolution: from a late entrant in DRAM manufacturing to a **stable and important contributor in the global memory supply chain**, supporting modern computing systems with reliable and essential memory technology.

3.3 Objectives

Nanya Technology is primarily focused on one clear mission: **to design and manufacture reliable DRAM memory that supports stable, efficient, and high-performance computing systems**. Instead of trying to cover multiple semiconductor areas, the company concentrates deeply on memory technology as its core strength.

A major objective for Nanya is **ensuring consistent and stable memory supply** for global customers. Since DRAM is a critical component in all computing devices, from laptops to servers, Nanya aims to maintain dependable production capacity so system manufacturers like HP Inc. can build and ship products without interruption.

Another key focus is **continuous improvement of DRAM technology**, especially as computing demands increase. This includes advancing memory standards such as DDR4 and DDR5, improving speed, reducing power consumption, and enhancing overall efficiency so that systems can handle modern workloads, including AI-driven applications powered by processors from Advanced Micro Devices.

Nanya also aims to **balance performance with cost efficiency**. In the highly cyclical memory industry, pricing and demand can change rapidly, so the company focuses on maintaining competitive production costs while still delivering quality and JEDEC-standard compliant products. This balance allows it to remain a stable supplier even during market fluctuations.

In addition, the company places strong importance on **supporting the broader electronics ecosystem**, including PC manufacturers, industrial systems, and consumer electronics. Its DRAM products are designed to integrate smoothly with storage solutions from companies like Kioxia, helping ensure overall system responsiveness and efficiency.

In simple terms, Nanya's objective is straightforward:

to provide dependable memory technology that keeps modern computers running smoothly, efficiently, and reliably across different types of devices and workloads.

3.4 Key Characteristics

Nanya Technology is best understood as a **focused and stability-driven memory manufacturer**, whose identity is built around reliability, consistency, and specialization in DRAM technology rather than broad semiconductor diversification.

At its core, Nanya is characterized by its **deep specialization in DRAM memory design and production**. Instead of spreading into CPUs, GPUs, or storage, the company concentrates entirely on dynamic random access memory. This narrow focus allows it to refine manufacturing processes and maintain steady expertise in DDR-based technologies used in modern computing systems.

Another defining trait is its **reliable and steady supply capability**. In the global PC ecosystem, memory supply stability is critical, and Nanya plays the role of a dependable Tier-2 supplier for manufacturers like HP Inc.. While not the largest producer, it is valued for its consistent output and ability to support system builders during periods of high demand or supply tightness.

Nanya is also known for its **cost-competitive and efficiency-oriented manufacturing approach**. Operating within the highly cyclical DRAM industry, the company focuses on controlling production costs while maintaining quality standards. This allows it to remain competitive even when market prices fluctuate significantly.

A further characteristic is its **strong alignment with industry memory standards (JEDEC compliance)**. This ensures that its DRAM products integrate seamlessly into global computing platforms, whether paired with processors from Advanced Micro Devices or storage systems from Kioxia.

Finally, Nanya is defined by its **incremental and stability-focused innovation style**. Rather than rapid disruptive shifts, it advances through steady improvements in DDR generations, power efficiency, and process refinement, ensuring long-term reliability across computing environments.

3.4 Products & Technologies

Nanya Technology focuses entirely on **DRAM (Dynamic Random Access Memory)** products, making it a specialized supplier of the short-term working memory used in almost all modern computing devices. Its product and technology strategy is centered

on delivering stable, efficient, and standards-compliant memory solutions that integrate smoothly into global PC platforms.

At the core of its portfolio are **DDR memory products**, especially DDR4 and DDR5 generations. These DRAM modules are widely used in laptops, desktops, and workstations because they directly impact system responsiveness, multitasking ability, and overall performance. In systems built by manufacturers like HP Inc., Nanya's memory acts as the "working space" that allows applications and operating systems to run smoothly alongside processors from Advanced Micro Devices.

Alongside standard DDR memory, Nanya also develops **low-power DRAM (LPDDR variants)** designed for mobile and energy-efficient systems. These products are optimized for reduced power consumption while maintaining adequate performance, making them suitable for thin laptops and portable devices where battery life is critical.

On the technology side, Nanya's strength lies in its **process engineering and DRAM fabrication capability**. The company designs and manufactures its own memory wafers using advanced semiconductor processes, ensuring control over quality, yield, and performance consistency. This vertical control allows it to maintain stable production across different DRAM generations.

Another important part of its technology portfolio is **JEDEC-standard compliance**, meaning all its memory products follow global industry specifications. This ensures compatibility across different platforms, whether paired with AMD processors or storage solutions from Kioxia, making integration into PC systems straightforward for OEMs.

In addition, Nanya continuously works on **incremental improvements in density, speed, and power efficiency**, rather than radical redesigns. This includes refining DDR architectures, improving data transfer rates, and optimizing energy usage to meet the growing demands of AI-driven and multitasking-heavy workloads.

In simple terms, Nanya's products and technologies can be summarized as:
a focused range of DRAM and low-power memory solutions designed to deliver

stable, efficient, and compatible system performance across modern computing platforms.

4.Kioxia(SSD / NAND Flash)

4.1 Introduction

Kioxia is a **semiconductor company that specializes in NAND flash memory**, which is a type of non-volatile storage technology used to permanently store data in electronic devices such as laptops, desktops, and servers. In simple terms, Kioxia provides the “long-term memory” of a computer—the part that keeps files, operating systems, and applications saved even when the device is turned off.

Unlike DRAM manufacturers such as Nanya Technology, which focus on temporary working memory, Kioxia focuses on **persistent storage solutions**. Its NAND flash chips are the core building blocks of Solid-State Drives (SSDs), which have replaced traditional hard drives in modern computing due to their much higher speed, durability, and efficiency.

Kioxia plays a key role in the PC ecosystem by supplying NAND flash memory that is integrated into SSDs used by system manufacturers like HP Inc.. These SSDs determine how fast a system boots, how quickly applications load, and how efficiently data is accessed and stored during everyday use.

At a technological level, Kioxia is known for developing advanced **3D NAND flash architecture**, where memory cells are stacked vertically to increase storage density while improving performance and reducing cost per bit. This innovation allows modern devices to store large amounts of data in smaller physical spaces without sacrificing speed.

In simple terms, Kioxia can be understood as the company that provides a computer with its **long-term digital storage system**, ensuring that data remains safe, accessible, and fast to retrieve whenever needed.

4.2 History

Kioxia has its origins in one of the most important inventions in modern computing: **NAND flash memory**, which was first developed in 1987 by Toshiba's semiconductor division. This breakthrough laid the foundation for modern solid-state storage, enabling data to be stored without moving parts.

For many years, the NAND flash business operated as part of Toshiba, where it grew into a global leader in memory innovation. During this period, Toshiba played a major role in advancing flash memory technologies, including early generations of NAND used in USB drives, SSDs, and mobile devices.

A major turning point came in **2018**, when Toshiba Memory Corporation was spun off as a separate entity to focus entirely on memory solutions. Shortly after, in **2019**, the company was officially rebranded as **Kioxia**, a name derived from the Japanese word "kioku" (memory) and the Greek word "axia" (value), symbolizing "memory value."

After becoming an independent company, Kioxia accelerated its focus on **3D NAND flash technology**, particularly its BiCS FLASH™ architecture. This innovation allowed memory cells to be stacked vertically, significantly increasing storage capacity and efficiency while reducing cost per bit.

In the early 2020s and into 2026, Kioxia strengthened its position as one of the world's leading NAND flash suppliers. Its products became widely used in **SSDs for laptops, desktops, and data centers**, including systems built by manufacturers like HP Inc.. At the same time, growing demand from AI computing and cloud infrastructure further increased the importance of high-performance NAND storage.

A notable recent development in its history is the **deepening collaboration across the memory ecosystem**, including strategic investments and partnerships with companies such as Nanya Technology, reflecting the increasing need for stable and scalable memory supply in the AI era.

Today, Kioxia's history represents a continuous evolution—from a division inside Toshiba to a **global independent leader in NAND flash innovation**, shaping the future of high-speed, high-density digital storage.

4.3 Objectives / Core Focus

Kioxia is primarily focused on one central mission: **to develop high-performance NAND flash memory that enables fast, reliable, and scalable digital storage for modern computing systems**. As a leading SSD and NAND manufacturer, its role is to ensure that data can be stored and accessed quickly, safely, and efficiently across everything from personal laptops to large-scale data centers.

A key objective for Kioxia is **advancing storage density and performance through continuous NAND innovation**. This includes improving its 3D NAND (BiCS FLASH™) technology by increasing the number of stacked memory layers, enhancing read/write speeds, and reducing power consumption. These improvements directly impact how quickly systems can boot, load applications, and handle large data workloads.

Another important goal is **supporting the growing demand for data-intensive and AI-driven computing**. As systems built by companies like HP Inc. become more powerful and AI-capable, storage requirements have increased significantly. Kioxia aims to provide SSD solutions that can keep up with high-speed data processing, especially in AI PCs, cloud computing, and enterprise servers.

Kioxia also focuses on **delivering reliability and endurance in storage systems**. Since NAND flash has a limited number of write cycles, the company works on improving durability, error correction technologies, and data retention quality. This ensures that stored information remains stable and secure over long periods of use.

In addition, the company is committed to **balancing cost efficiency with technological advancement**. By optimizing manufacturing processes and improving production yields, Kioxia aims to make high-capacity SSDs more affordable while still maintaining performance standards required by modern systems.

Finally, Kioxia plays a broader role in **strengthening the global memory and storage ecosystem**, working alongside DRAM suppliers like Nanya Technology and processor platforms from Advanced Micro Devices to ensure complete system-level performance.

4.4 Key Characteristics

Kioxia is best understood as a **technology-focused and innovation-driven storage company**, whose identity is built around high-density NAND flash development, reliability, and large-scale data storage performance.

At the core of its characteristics is its **leadership in NAND flash innovation**, particularly through its BiCS FLASH™ 3D NAND architecture. Instead of storing data in a flat structure, Kioxia stacks memory cells vertically, which allows significantly higher storage capacity within the same physical space. This design approach has become a foundation for modern SSD technology used in laptops and enterprise systems.

Another defining characteristic is its **focus on high-density and scalable storage solutions**. Kioxia designs its products to support increasing global data demands, especially from AI computing, cloud infrastructure, and high-performance PCs. This makes its technology essential in systems built by manufacturers like HP Inc., where fast and large-capacity SSDs are required.

Kioxia is also known for its **strong emphasis on performance efficiency**, balancing speed, endurance, and power consumption. Its NAND flash products are optimized to deliver fast read and write speeds while maintaining stability over long usage cycles, which is critical for both consumer and enterprise workloads.

Another key trait is its **reliability and data integrity focus**. Since NAND flash is used for permanent data storage, Kioxia invests heavily in error correction, endurance improvement, and data retention technologies to ensure stored information remains safe and accessible over time.

The company also stands out for its **deep specialization in storage rather than diversification**. Unlike broader semiconductor firms, Kioxia concentrates almost entirely on NAND flash and SSD technologies, allowing it to build strong expertise and continuous improvement in this specific field.

Finally, Kioxia is characterized by its **role in the global memory ecosystem**, working alongside DRAM suppliers like Nanya Technology and processor designers such as

Advanced Micro Devices to ensure complete system-level performance in modern computing devices.

4.5 Products & Technologies

Kioxia is a **pure-play NAND flash and SSD company**, meaning its entire product and technology ecosystem is built around one core area: **non-volatile storage that retains data even when power is turned off**. Its technologies are designed to deliver high-speed, high-capacity, and reliable storage for everything from personal laptops to large-scale cloud systems.

At the center of its product lineup is **NAND flash memory**, which is the fundamental building block of all its storage solutions. This memory is used to create Solid-State Drives (SSDs), which replace traditional hard drives by offering much faster data access, lower power consumption, and greater durability. These SSDs are widely used in systems built by manufacturers such as HP Inc..

A key technology developed by Kioxia is **BiCS FLASH™ (Bit Cost Scaling) 3D NAND architecture**. Instead of storing memory cells on a single flat layer, BiCS FLASH stacks them vertically in multiple layers. This structure significantly increases storage density, improves performance, and reduces cost per gigabyte, making it possible to build compact SSDs with large capacities.

Within this 3D NAND technology, Kioxia produces different types of memory cells such as **TLC (Triple-Level Cell) and QLC (Quad-Level Cell)**. These variants balance performance, endurance, and cost depending on the application—TLC for higher performance SSDs and QLC for high-capacity, cost-efficient storage solutions.

Another important part of its technology portfolio is **client and enterprise SSD solutions**. Client SSDs are used in laptops and desktops for fast boot times and application loading, while enterprise SSDs are designed for data centers, cloud computing, and AI workloads that require extremely high reliability and sustained performance.

Kioxia also invests heavily in **advanced controller integration and firmware optimization**. These technologies manage how data is written, stored, and retrieved

from NAND flash, ensuring stability, error correction, and long-term durability of stored data.

In addition, the company continuously develops **high-layer 3D NAND scaling technologies**, increasing the number of stacked layers in memory chips to push storage density even further while maintaining performance efficiency.

Supply Chain Relationship Between AMD, HP, Nanya Technology, and Kioxia

The relationship between Advanced Micro Devices, HP Inc., Nanya Technology, and Kioxia can be understood as a **layered PC supply chain**, where each company contributes a different but interconnected part of a complete computing system.

1. Component Suppliers (Core Hardware Layer)

At the foundation are the semiconductor and memory manufacturers:

- **AMD** provides the **processor (CPU/APU)**, which acts as the main computing engine of the system. It handles calculations, system control, and increasingly AI processing through built-in NPU capabilities.
- **Nanya Technology** supplies **DRAM (RAM)**, which serves as the short-term working memory that supports multitasking and system responsiveness.
- **Kioxia** provides **NAND flash memory**, which is used in SSDs for long-term storage of data, applications, and operating systems.

Together, these three form the **core internal hardware stack** of a computer: processing, working memory, and permanent storage.

2. System Integrator (Product Assembly Layer)

Above these component suppliers is **HP Inc.**, which acts as the **Computer System Manufacturer (CSM/OEM)**. HP does not manufacture the chips itself; instead, it:

- Selects and sources CPUs, DRAM, and SSD components
- Designs motherboards and system architecture

- Integrates hardware into finished laptops and desktops
- Adds firmware, BIOS, and system-level optimization
- Performs testing, validation, and quality control

This makes HP the company responsible for turning individual components into a **complete, usable product**.

3. How They Work Together in a Laptop

Inside a typical HP laptop:

- **AMD CPU** performs processing and AI tasks
- **Nanya DRAM** provides temporary working space for active tasks
- **Kioxia SSD** stores the operating system, files, and applications
- **HP system design** ensures all components communicate efficiently and operate reliably

This integration creates a balanced system where performance depends on all layers working together, not just a single component.

4. Relationship Nature (How They Connect)

- **AMD, Nanya, and Kioxia** = upstream component suppliers
- **HP** = downstream system builder and integrator
- All are connected through **qualification, compatibility testing, and supply agreements**
- Their collaboration ensures performance, cost balance, and product reliability in final devices

5. Market Context & Industry Trends

The 2026 technology landscape for companies like Advanced Micro Devices, HP Inc., Nanya Technology, and Kioxia is being shaped by a major transition toward **AI-driven computing and a tightly constrained memory supply chain**.

1. Shift Toward AI PCs and Intelligent Computing

One of the biggest trends in 2026 is the rapid adoption of **AI PCs**. Modern systems are no longer just designed for traditional workloads; they now integrate on-device AI capabilities through processors like AMD Ryzen AI.

Businesses are increasingly deploying AI PCs to improve productivity, automate tasks, and reduce reliance on cloud processing. As a result, PCs are evolving into **AI-enabled productivity systems**, not just general-purpose computers.

2. Strong Growth in AI-Driven Memory Demand

AI infrastructure is creating a major surge in demand for both DRAM and NAND flash. Data centers, AI servers, and enterprise systems require significantly more memory per device, especially high-performance DRAM and SSD storage.

This is leading to a structural “memory supercycle,” where demand consistently exceeds supply, particularly in advanced memory segments used for AI workloads.

3. Tight Supply and Rising Memory Prices

A key industry condition in 2026 is **persistent memory shortage**. DRAM and NAND supply remains tight because manufacturers are prioritizing high-margin products such as AI server memory and enterprise SSDs.

This has led to strong price increases across the ecosystem:

- DRAM prices rising sharply due to limited capacity
- NAND flash prices increasing as enterprise demand grows
- SSD costs remaining elevated due to constrained supply

Overall, pricing is being driven more by supply limitations than by traditional consumer demand cycles.

4. Structural Shift in DRAM and NAND Allocation

Memory manufacturers are reallocating production capacity toward **AI servers and cloud infrastructure**, which offer higher profitability. As a result:

- Less capacity is available for consumer PCs
- DDR5 adoption is accelerating while DDR4 supply tightens
- Enterprise-grade memory is prioritized over mainstream segments

This shift is fundamentally changing how memory is distributed across industries.

5. Moderate Growth in PC Shipments but Higher Value Systems

While PC shipment growth remains modest, the value per device is increasing. Consumers and enterprises are shifting toward higher-end systems with:

- Faster processors (AMD Ryzen AI platforms)
- More DRAM capacity (Nanya-supported memory configurations)
- Faster SSD storage (Kioxia NAND-based solutions)

This means the market is moving from **volume-driven growth to value-driven growth**.

6. Increasing Importance of Supply Chain Collaboration

Because of tight supply conditions, coordination between companies is becoming more important. HP depends heavily on:

- AMD for processing performance
- Nanya for stable DRAM supply
- Kioxia for SSD storage capacity

This interconnected supply chain is becoming more strategic, with long-term partnerships and investment agreements becoming more common to secure supply stability.

6.Risks & Challenges

The ecosystem involving Advanced Micro Devices, HP Inc., Nanya Technology, and Kioxia operates in a fast-changing and highly interconnected industry. While the growth outlook is strong due to AI-driven computing, several structural risks and challenges shape how these companies operate in 2026.

1. Memory Supply Volatility (DRAM & NAND Cycles)

One of the most persistent challenges is the **cyclical nature of memory markets**. DRAM and NAND prices often fluctuate sharply based on supply-demand balance. When demand rises—especially from AI servers—supply can become constrained, leading to price spikes. When demand slows, oversupply can quickly push prices down.

This volatility directly affects:

- Nanya (DRAM production stability)
- Kioxia (NAND supply planning)
- HP (cost of finished systems)

2. AI-Driven Supply Pressure

The rapid expansion of AI infrastructure is creating **unprecedented demand for high-performance memory and storage**. Cloud providers and data centers often receive priority allocation, which can limit availability for PC manufacturers.

As a result:

- PC supply chains may face component shortages
- Cost of DRAM and SSDs may remain elevated
- System pricing becomes harder to stabilize

3. Geopolitical and Regional Manufacturing Risks

The semiconductor supply chain is heavily concentrated in Asia, particularly Taiwan, Japan, and South Korea. This creates exposure to:

- Geopolitical tensions
- Trade restrictions or export controls
- Regional disruptions (natural disasters, logistics delays)

Since Nanya and Kioxia are based in Taiwan and Japan respectively, and AMD and HP rely on global manufacturing networks, any disruption can ripple across the entire ecosystem.

4. High Competition and Margin Pressure

All four companies operate in highly competitive markets:

- AMD competes with other CPU and AI chip providers
- HP competes with global PC vendors
- Nanya competes with larger DRAM manufacturers
- Kioxia competes in NAND flash against other global memory leaders

This competition leads to **pricing pressure**, especially in consumer markets where margins are thin.

5. Technology Transition Complexity

The industry is undergoing rapid transitions:

- DDR4 → DDR5 in memory
- PCIe Gen4 → Gen5 SSDs
- AI integration at hardware level

These transitions require:

- New manufacturing processes
- System redesign by OEMs like HP
- Compatibility testing across all components

Delays or inefficiencies in adoption can create bottlenecks.

6. Cost and Power Efficiency Balancing

As performance increases, so does power consumption and thermal demand. System builders like HP must constantly balance:

- High-performance processors (AMD)
- High-speed memory (Nanya)
- Fast SSD storage (Kioxia)
- Battery life and heat management

This balance becomes more difficult in thin laptops and AI PCs.

7. Demand Uncertainty in Consumer Markets

While enterprise and AI demand is strong, **consumer PC demand remains uneven**. Factors such as economic conditions, upgrade cycles, and regional demand shifts can impact overall PC shipment stability.

7.Sustainability & Future Outlook

The 2026 outlook for the ecosystem involving Advanced Micro Devices, HP Inc., Nanya Technology, and Kioxia is shaped by two parallel forces: **increasing focus on sustainability** and **rapid evolution toward AI-driven, high-efficiency computing**.

1. Energy Efficiency as a Core Design Priority

A major sustainability direction across the entire supply chain is reducing energy consumption at every level of computing.

- AMD is improving power efficiency in CPUs and AI-enabled processors
- HP is optimizing system-level power management in laptops and desktops
- Nanya is developing lower-power DRAM technologies

- Kioxia is improving NAND efficiency to reduce SSD energy usage

The shared goal is to deliver **higher performance per watt**, which is becoming more important than raw performance alone.

2. Sustainable Manufacturing and Responsible Production

All companies in the ecosystem are increasingly focused on reducing the environmental impact of semiconductor production.

This includes:

- Lowering carbon emissions in fabrication plants
- Reducing water and chemical usage in chip manufacturing
- Improving yield efficiency to reduce electronic waste
- Increasing use of recyclable and eco-friendly materials in system design

HP, in particular, emphasizes **circular economy principles**, such as recyclable laptop materials and reduced packaging waste.

3. Transition Toward AI-Optimized Computing Platforms

The future of the industry is strongly tied to **AI integration at the hardware level**.

- AMD is embedding NPUs directly into processors for local AI processing
- HP is designing AI-ready PCs for productivity and enterprise use
- Nanya is ensuring memory bandwidth supports AI workloads
- Kioxia is scaling SSD performance for large AI data requirements

This shift reduces reliance on cloud computing for many tasks, improving both **latency and energy efficiency**.

4. Smarter and More Integrated Supply Chains

The ecosystem is moving toward tighter collaboration between component suppliers and system builders.

Instead of operating independently:

- AMD, Nanya, and Kioxia increasingly align with HP's system requirements
- Long-term supply agreements and strategic partnerships are becoming more common
- Inventory and production planning are being optimized using data-driven forecasting

This creates a more **resilient and predictable supply chain**, especially important during memory shortages.

5. Continued Growth of High-Value Computing

While overall PC shipment growth is moderate, system value is expected to rise.

Future systems will feature:

- More advanced CPUs with AI capabilities
- Higher DRAM capacities for multitasking and AI workloads
- Faster SSD storage with larger capacities
- Better integration between CPU, memory, and storage

This means the industry is shifting from **mass volume growth to high-performance value growth**.

6. Long-Term Industry Direction

The long-term direction of the ecosystem is becoming clear:

- PCs are evolving into **AI-native computing devices**
- Memory and storage are becoming more strategically important than ever
- System integration is becoming the key differentiator (HP's role)
- Efficiency, not just performance, defines next-generation computing

Conclusion

The ecosystem formed by Advanced Micro Devices, HP Inc., Nanya Technology, and Kioxia represents a complete and interdependent structure that powers modern computing devices.

Each company plays a clearly defined role: AMD provides the processing and AI compute engine, Nanya supplies the working memory that enables multitasking and system responsiveness, Kioxia delivers fast and reliable long-term storage through SSD technology, and HP integrates all of these components into finished, user-ready systems. Together, they form a layered value chain where performance depends on coordination rather than any single company alone.

In the context of 2026, this ecosystem is undergoing a major transformation driven by **AI adoption, tighter memory supply, and increasing demand for energy-efficient computing**. Systems are becoming more intelligent, more integrated, and more dependent on close collaboration between hardware suppliers and system manufacturers.

At the same time, challenges such as supply volatility, geopolitical risks, and rapid technology transitions continue to shape how these companies operate. Despite these pressures, the direction of the industry remains clear: **higher-value computing systems with stronger AI capabilities, better efficiency, and deeper hardware integration**.

In simple terms, this ecosystem shows how modern computers are no longer built by a single company, but by a **carefully connected network of specialized innovators working together to deliver faster, smarter, and more efficient computing experiences**.

REFERENCE

AMD: <https://www.amd.com/>

HP Inc.: <https://www.hp.com/>

Nanya Technology: <https://www.nanya.com/>

Kioxia: <https://www.kioxia.com/>Oracle Corporation. (2026).